

NETWORK TRAFFIC DATASET ANALYSIS

Video Streaming (TCP) vs VoIP Call (UDP).

1. Introduction

This project compares two types of network communication traffic: Video_Streaming TCP and VoIP_Call UDP. The purpose is to understand their packet-length characteristics and use the results to recommend a suitable priority policy during network congestion.

2. Problem Statement

A company uses its network for video meetings and voice calls. Network engineers need to understand how these applications behave differently so that important real-time traffic can be protected when network capacity becomes limited. This project uses Activity_Type, Protocol, and Length to separate and compare the two traffic types.

3. Objectives

- Separate Video_Streaming TCP and VoIP_Call UDP traffic.
- Calculate basic packet-length statistics.
- Compare the two communication patterns using charts.
- Identify the major differences in packet-size characteristics.
- Recommend which traffic should receive higher priority during congestion.

4. Dataset Description

The uploaded source file is a 1,951-page PDF representation of the network traffic dataset. Its table uses the columns Time, Protocol, Length, and Activity_Type. The source contains Video_Streaming TCP records and VoIP_Call UDP records. For example, Video_Streaming records include repeated TCP packets of 1514 bytes, while VoIP_Call records include UDP packets such as 173, 125, 277, 254 and other smaller values.

Source evidence from the uploaded file: the table header is shown at the beginning of the file, and Video_Streaming TCP rows and VoIP_Call UDP rows are present throughout the dataset.

Column	Meaning	Used for
Time	Packet timestamp	Traffic sequence
Protocol	Network protocol	TCP/UDP separation
Length	Packet size in bytes	Statistical comparison
Activity_Type	Application category	Video/VoIP separation

5. Data Selection and Method

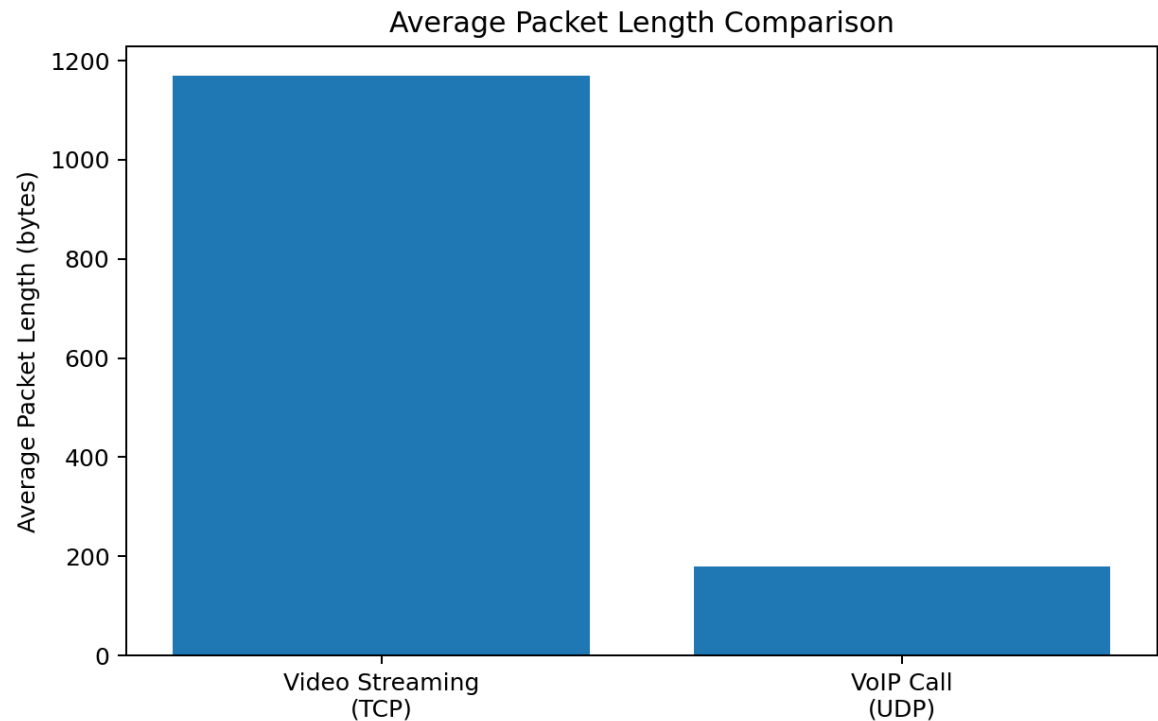
- 1. Extract the table text from the uploaded PDF.
- 2. Select rows where Activity_Type = Video_Streaming and Protocol = TCP.
- 3. Select rows where Activity_Type = VoIP_Call and Protocol = UDP.
- 4. Use Length as the packet-length variable.
- 5. Calculate count, mean, median, minimum, maximum, and standard deviation.
- 6. Visualize the mean and the packet-length distributions.

6. Statistical Results

Statistic	Video Streaming (TCP)	VoIP Call (UDP)
Packets	41,079	11,562
Average	1,170.04	179.68
Median	1,514.00	181.00
Minimum	54.00	60.00
Maximum	1,514.00	480.00
Std Dev	618.34	84.86

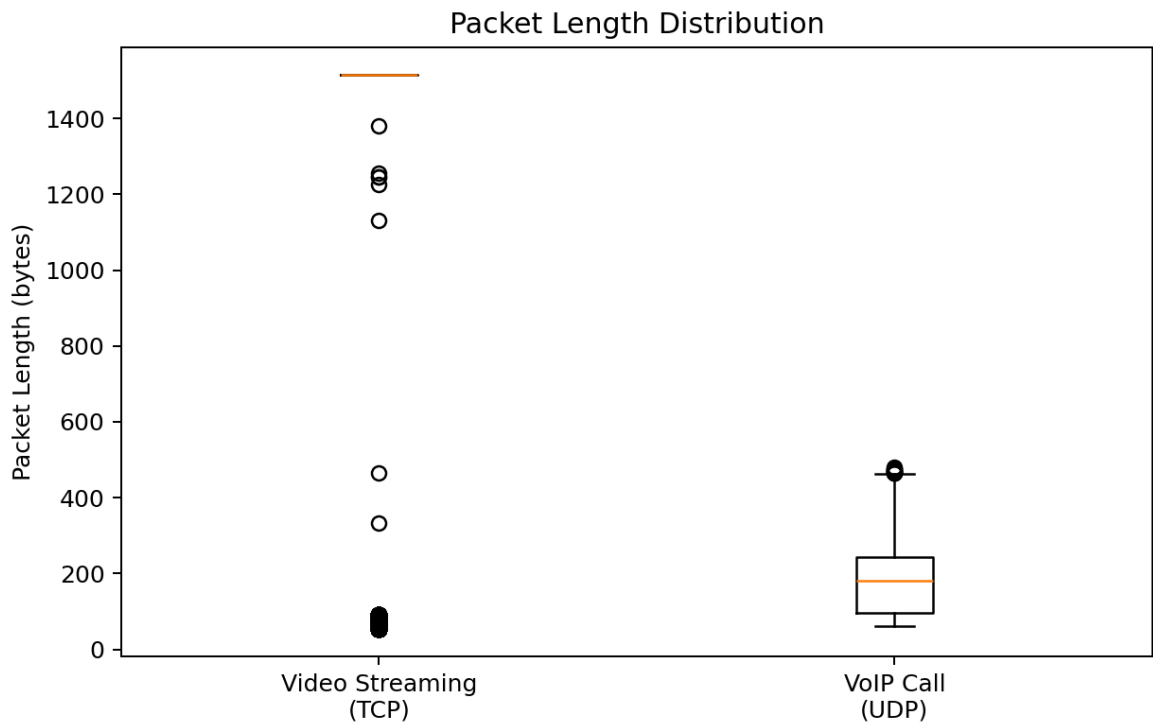
Video Streaming TCP contains 41,079 selected packets, while VoIP UDP contains 11,562 selected packets. The mean packet length is 1170.04 bytes for Video Streaming and 179.68 bytes for VoIP.

7. Bar Chart – Average Packet Length



The chart compares the average packet length of the two traffic classes.

8. Box Plot – Packet Length Distribution



The box plot shows the median and spread of packet lengths. Video Streaming shows many large packets, including repeated 1514-byte TCP packets, whereas VoIP traffic is concentrated in smaller packet sizes.

9. Observations

- Video Streaming TCP average packet length = 1170.04 bytes.
- VoIP Call UDP average packet length = 179.68 bytes.
- Video Streaming median packet length = 1514.00 bytes.
- VoIP Call median packet length = 181.00 bytes.
- Video Streaming maximum packet length = 1,514 bytes.
- VoIP Call maximum packet length = 480 bytes.
- The Video_Streaming traffic contains frequent large TCP packets and therefore represents a higher-volume data transfer pattern.
- The VoIP_Call traffic contains smaller, recurring UDP packets and represents a more regular real-time communication pattern.

10. Communication Characteristics

Characteristic	Video Streaming (TCP)	VoIP Call (UDP)
Packet size	Generally larger	Generally smaller
Traffic nature	High-volume / bursty	Regular / recurring
Protocol	TCP	UDP
Delay sensitivity	Moderate; buffering may help	High; real-time conversation
Congestion effect	Buffering and reduced quality	Delay, jitter and interruptions

11. Priority Recommendation During Congestion

VoIP UDP traffic should receive HIGHER PRIORITY during network congestion.

Reason: voice communication is interactive and highly sensitive to delay, jitter, and packet loss. If voice packets are delayed, the conversation can become difficult to understand. Video traffic can often tolerate some delay because buffering can temporarily absorb variation. Therefore, a network QoS policy should place VoIP in a low-latency, high-priority queue and allow Video Streaming to use the remaining capacity.

The recommendation is based on the observed traffic characteristics as well as the real-time nature of VoIP. Packet size alone is not the reason for the priority decision.

12. Suggested Network QoS Policy

- Classify VoIP UDP traffic as high priority.
- Use a low-latency queue for VoIP.
- Give Video Streaming TCP a lower or standard priority during congestion.
- Monitor delay, jitter, packet loss, and throughput after applying QoS.
- Adjust queue weights according to business requirements and measured network performance.

13. Conclusion

The analysis shows that Video_Streaming TCP and VoIP_Call UDP have different communication characteristics. Video Streaming has an average packet length of 1170.04 bytes, while VoIP averages 179.68 bytes. The dataset also shows frequent large TCP packets for Video Streaming and smaller recurring UDP packets for VoIP. Because VoIP is a real-time service with strong sensitivity to delay and jitter, VoIP traffic should receive higher priority during network congestion. This approach protects voice quality while allowing video traffic to continue using the remaining network capacity.

14. Tools Used

- Python
- PDF text extraction
- Statistical analysis
- Matplotlib charts
- Microsoft Word-compatible DOCX

15. Reference

Source file provided for the project: network_traffic_dataset.pdf.

Kaggle dataset link supplied in the project question:

<https://www.kaggle.com/datasets/sxham04/network-traffic-dataset/data>